

4 ADDERS - SUBTRACTORS

4.1 Theoretical part

4.1.1 Half-adder

A half-adder is called a combinational circuit that will have as its inputs two binary digits and will output their sum and the possible prisoner. If X, Y are the two single-digit binary numbers that will be added together and S, C their sum and the possible detainer respectively then the following truth table applies:

X	Y	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

The truth table yields the following logical equations:
for the sum and the retainer:

$$S = \bar{X}Y + X\bar{Y} = X \oplus Y$$

$$C = XY$$

The logic circuit is implemented as follows (figure 1).

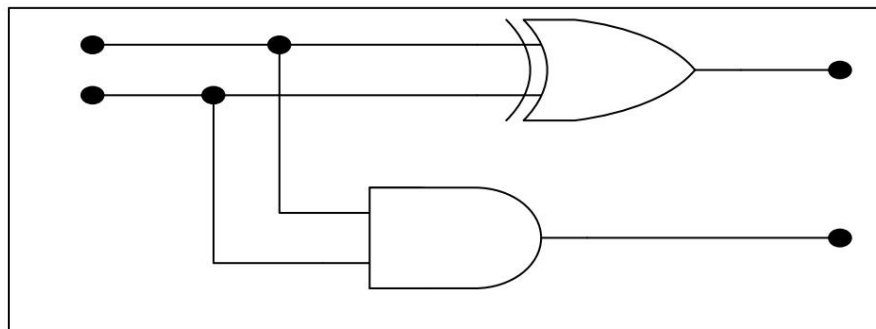


Fig. 1 Half-adder logic circuit

4.1.2 Full adder

To add two binary numbers with more than one digits each, in the circuit that will calculate the sum of the digits of each column, we should also take into account the prisoner who has probably resulting from the sum of the digits of the previous column.

If X_n , Y_n are the digits of the "n" column and C_{n-1} is the possible held by the sum of the digits of the previous column then S_n we define the sum of X_n , Y_n and C_{n-1} and as C_n the new possible. Thus the truth table of the full adder has three inputs X_n , Y_n and C_{n-1} and two outputs S_n and C_n and is as follows:

X_n	Y_n	C_{n-1}	S_n	C_n
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The logical equations resulting from the truth table for the S_n , C_n are as follows:

$$S_n = \overline{X_n} \overline{Y_n} C_{n-1} + \overline{X_n} Y_n \overline{C_{n-1}} + X_n \overline{Y_n} \overline{C_{n-1}} + X_n Y_n C_{n-1}$$

$$S_n = (\overline{X_n} \overline{Y_n} X_n Y_n) C_{n-1} + (\overline{X_n} Y_n X_n Y_n) \overline{C_{n-1}}$$

$$S_n = (X_n Y_n) C_{n-1} + (X_n Y_n) \overline{C_{n-1}}$$

$$S_n = X_n Y_n C_{n-1} + (X_n Y_n) \overline{C_{n-1}} \quad (1)$$

$$C_n = \overline{X_n} Y_n C_{n-1} + X_n \overline{Y_n} C_{n-1} + X_n Y_n \overline{C_{n-1}} + X_n Y_n C_{n-1}$$

$$C_n = (\overline{X_n} Y_n X_n Y_n) C_{n-1} + (\overline{C_{n-1}} - 1 C_{n-1} - 1) X_n Y_n$$

$$C_n = (X_n Y_n) C_{n-1} + X_n Y_n \quad (2)$$

Note: The Karnaugh table does not help in simplification – do not giving adjacent ones or zeros - in this case as well as below in the complete subtractor and thus we do algebraic simplification.

From equations (1) and (2) we design the logic circuit of full adder (figure 2).

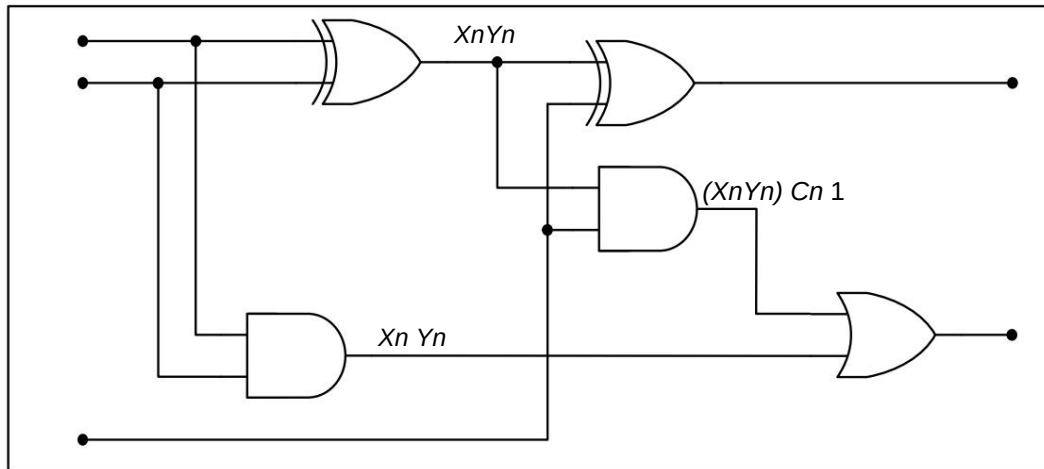


Fig. 2. Full adder logic circuit.

4.1.3 Semi-subtractor

The semi-subtractor is used to subtract single digits. binary numbers. It has two inputs X, Y and outputs the difference D and the borrowed B that possibly arises. The truth table for the semi-subtractor shown below:

X	Y	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

From the truth table, the logical equations for the difference and the borrowed amount given below:

$$D = XY \oplus \bar{X}Y = X \oplus Y$$

$$B = \bar{X}Y$$

From the equations we draw the half-subtractor circuit (figure 3).

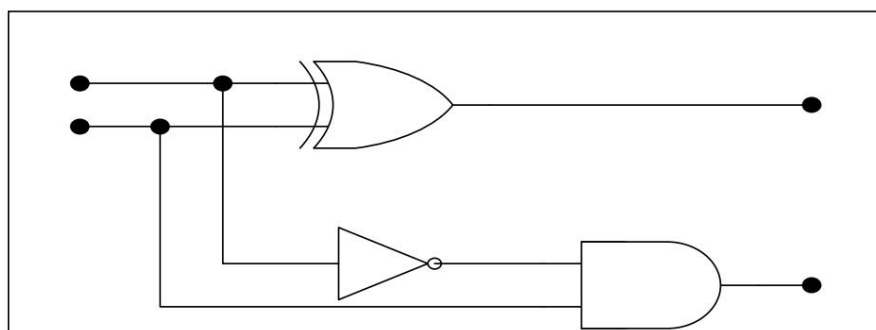


Fig. 3 Half-subtractor logic circuit

4.1.4 Complete subtractor

Analogously to the full adder, the full subtractor circuit should also be included for the difference in digits of a column possible borrowed word resulting from subtracting its digits previous column.

The truth table below for the complete subtractor will have three inputs X_n , Y_n and B_{n-1} , where X_n , Y_n the 'n' digit of the minuend and the respectively deductible and B_{n-1} the possible borrower from the previous subtraction and two outputs D_n and B_n , where D_n and B_n are the difference and potential borrower respectively resulting from the deduction of of a specific 'n' column.

X_n	Y_n	B_{n-1}	D_n	B_n
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

The logical equations for the difference and the borrower are as follows:

$$D_n = \overline{X_n} \overline{Y_n} \overline{B_{n-1}} + \overline{X_n} Y_n \overline{B_{n-1}} + X_n \overline{Y_n} \overline{B_{n-1}} + X_n Y_n \overline{B_{n-1}}$$

$$D_n = (\overline{X_n} \overline{Y_n} X_n Y_n) \overline{B_{n-1}} + (\overline{X_n} Y_n X_n \overline{Y_n}) \overline{B_{n-1}}$$

$$D_n = (X_n Y_n) \overline{B_{n-1}} + (\overline{X_n} \overline{Y_n}) B_{n-1}$$

$$D_n = X_n Y_n B_{n-1} - (3) \quad 1$$

$$\begin{array}{ccccccc} _ & _ & & _ & & _ & \\ _ & _ & & _ & & _ & \\ \hline & & & & & & \end{array} \quad (4)$$

The circuit of the complete subtractor is as follows (figure 4).

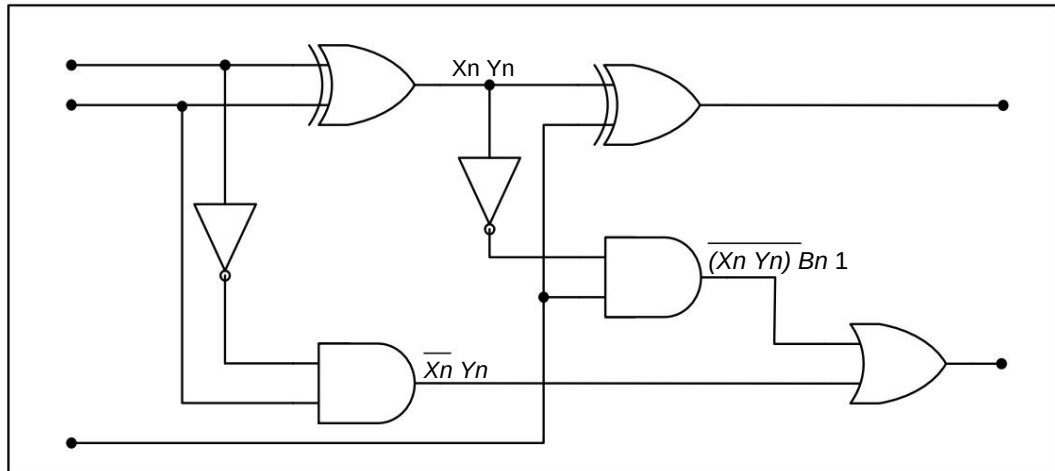


Fig. 4 Complete subtractor circuit.

4.1.5 The integrated 7483

The 7483 integrated circuit is an adder of two binary numbers. four-bit whose terminals are shown in figure 5. We observe that it has the following inputs:

A4 A3 A2 A1 one additional
 B4 B3 B2 B1 the other one added
 C0 potential entry detainee

and the following outputs:

S4 S3 S2 S1 the sum
 C4 the potential prisoner of exit

(Input latch C0 will be present in the case of addition higher order digits, so C0 is connected to the output carry C4 of the lowest order digits).

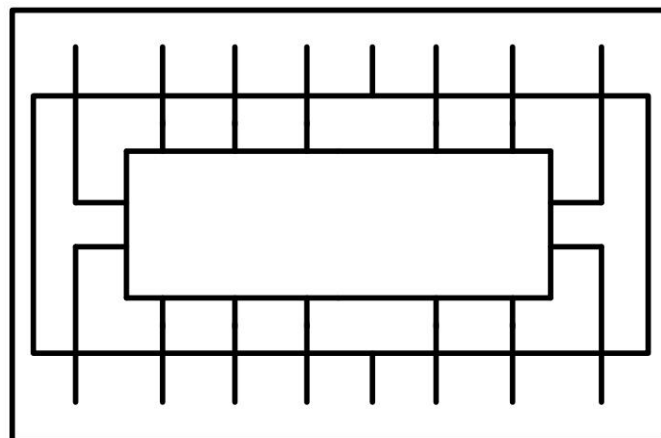


Fig. 5. Terminals of the 7483 integrated circuit.

Addition of 4bit numbers

The two numbers $A_4 A_3 A_2 A_1$ and $B_4 B_3 B_2 B_1$ are added together with any input hold C_0 and we have as outputs the sum $S_4 S_3 S_2 S_1$ and the exit prisoner C_4 .

That is:

$$\begin{array}{r}
 A_4 A_3 A_2 A_1 \\
 + B_4 B_3 B_2 B_1 \\
 + \quad \quad \quad C_0 \\
 \hline
 \text{Sum} \quad \quad = S_4 S_3 S_2 S_1
 \end{array}$$

and output holdout C_4 , the sum is the number:
 $C_4 S_4 S_3 S_2 S_1$

Subtracting 4bit numbers

We recall that the subtraction of two numbers $A-B$ can be transformed in addition if we add to the subtracted the complement of 2 of subtracted and ignore the output retainer, which is a unit. These apply to the case where $A \geq B$, and the expected result is positive or zero.

In the case where $A < B$ then the output hold is 0, which means that the result is negative, while the result appears in the form of a complement to 2.

Therefore:

If $A \geq B$

The output hold is 1 indicating that the difference is positive number and is ignored from the result so:

$$A - B = A + B_{\text{complement}2}$$

In the case where $A = B$ the holdend is 1 and the result is 0.

e.g. 1001 (9)	1001
-0111 (7)	+1001
	10010

Inmate 1 ignored positive result 0010 (2).

If $A < B$

The output retain is 0, so we have a negative result which must be completed in terms of 2. So:

$$A - B = -[A + B_{\text{2's complement}}]_{\text{2's complement}}$$

B.C	1000 (8)	1000
	-1011 (11)	+0101
		1101

Holding 0 result: $-\text{[1101 complement 2]} = -0011 (-3)$

To implement an adder-subtractor circuit with the 7483 integrated circuit, we should also know that: The 2's complement is obtained from the 1's

complement if we add 1 to it. Four EX-OR gates can give us either a 4-bit number that they accept at their inputs or its

1's complement controlled by a common input (Mode). This is because from the truth table of the EX-OR we have:

X 1 — and X 0 X

Based on the above, we design an addition and subtraction circuit of 4-bit numbers with the 7483 and 7486 integrated circuits (EX-OR) (figure 6).

We observe that during the subtraction with $M=1$ we also have $C0=1$ and thus the 2's complement of the B number results from the 1's complement through the EX-OR and then the addition of one to the input hold $C0$.

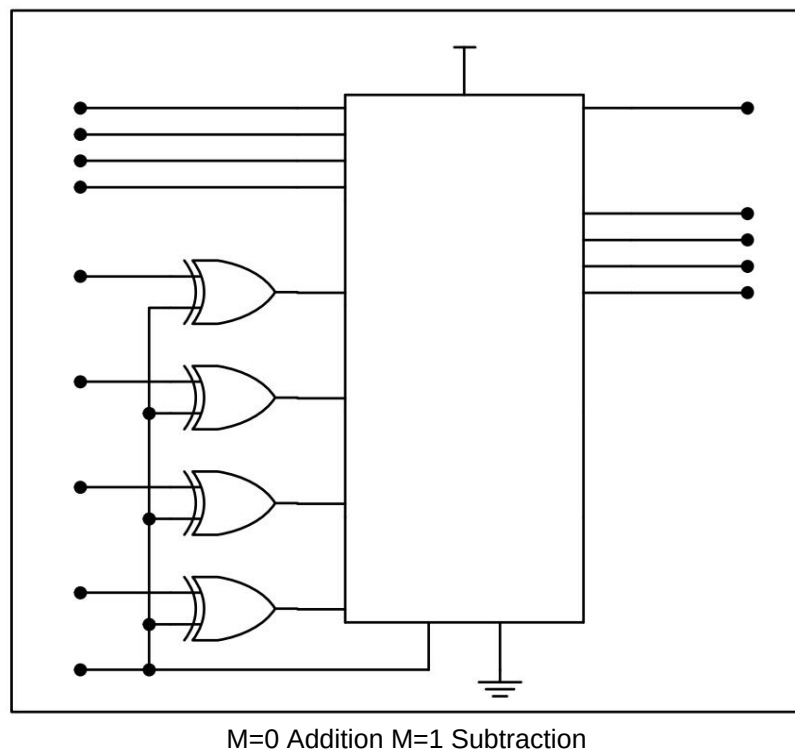


Fig. 6 Circuit for adding and subtracting 4-bit numbers with the integrated circuits 7486, 7483

4.2 Work Execution

1) Connect the half-adder circuit (figure 1) and verify the truth table for all combinations of income X and Y.

2) Convert the half-adder circuit into a half-subtractor circuit (figure 3) adding the inverter and verify the truth table for the half-subtractor for all combinations of inputs X and Y.

3) Connect a full adder circuit (figure 2) and verify the truth table for all combinations of X_n C_{n-1} , Y_n and

4) By adding two inverters, you convert the circuit into a complete adder in a full subtractor circuit (figure 4) and verify the truth table of the complete subtractor for all combinations of X_n Y_n and B_{n-1} ,

5) Connect the circuit in figure 6. Connect the outputs S_4 , S_3 , S_2 , S_1 and C_4 to LED, the inputs of number B to switches 0 or 5V and the inputs of number A to 0 or 5V forming the number 1001.

a) Connect input M to 0V for addition mode. Check the its operation in addition by performing the additions:

A 1001 +0110	1001	1001	1001
B	+1000	+0100	+1100

b) Connect the M input to 5V for subtraction mode. Check the its function in subtraction by performing the subtractions:

A 1001 1001 1001 1001
B -0111 -1111 -0100 -1011

6) With the addition of two EX-OR gates and the property of invert one of their inputs when the other is at logic '1' combine the full adder and full subtractor circuits to design a full addition-subtraction circuit where the choice of operation will be done with a Mode control input.

7) Figure 7 below shows a 4-bit parallel adder. It consists of one half adder (HA) and three full adders (FA).

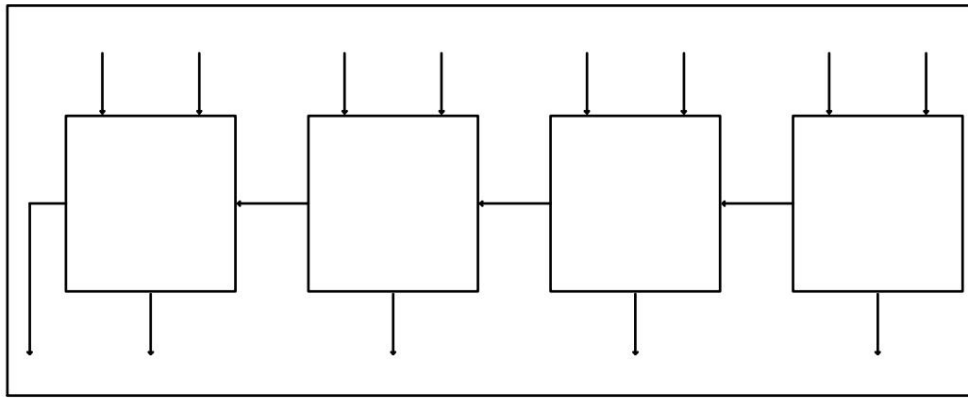


Fig. 7. 4-bit parallel adder.

- a) Why is HA used in the addition of lower-order digits?
- b) However, if instead of HA we had FA there, design a circuit with two such parallel 4-bit adders that would add two 8-bit numbers.
- c) In figure 7, the numbers 1001 and 1011 are added. Complete the values of S_0, S_1, S_2, S_3 and CO, C_1, C_2, C_3 .
- d) If the half-adder circuit (figure 1) is represented by a block diagram as in figure 8, show by carefully observing figure 2 of the full adder that the full adder can be implemented by two HAs and an OR gate. Draw a block diagram of the full adder in this way.

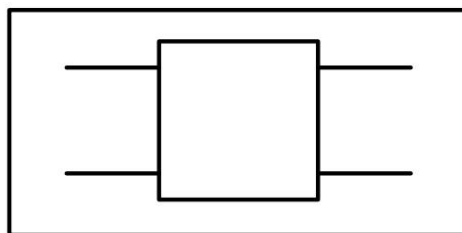


Fig.8 Block diagram of a half-adder

4.3 Construction

1's complement addition and subtraction circuit

